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Data Analytics Group 11



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Applying Data Analytics Principles for Human Resource Analytics: Employee Turnover Prediction and Workforce Segmentation

The studies focus on business related problems, such as factors affecting the working state of the workers and ultimately predicting employee attrition. In this work, Data Analytics study is applied in Human Resource Management (HRM) with the dataset which is the personnel record of a global company. The models employed were two machine learning classifiers; namely, Random Forest Classifier and Decision Tree Classifier for predicting employment status [17] and [22]. The overall accuracy of the Random Forest Classifier was 54% and feature significance was 95.8% for Salary_INR and F1-Score of the Active class was 0.70. The Decision Tree Classifier performed an overall accuracy of 69% with feature significance of Salary_INR as 61.0%, F1-Score as 0.81 for Active class. The findings offered some suggestions to the organisation to decrease employee turnover.

Keywords—HR Analytics, Employee Attrition, Random Forest Classification, Decision Tree Classification, Feature Importance, Predictive Modeling.

I. INTRODUCTION

Analytics for human resources (HR) is an important field in contemporary businesses. Converting a routine process into a data-driven, strategic decision-making engine [10], [11]. Due to recruiting costs and a loss of organizational momentum, large international firms often experience financial and operational losses as a result of staff turnover [14]. Conventional HR procedures, such as examining departure interviews, are based on a more reactive structure. Predictive data analytics, on the other hand, enables firms to assess compensation distribution and detect turnover concerns, which should have a positive effect on organizational performance [9], [15]. HR analytics integration must, however, also be able to handle moral dilemmas pertaining to the administration of employee data [5].

Using a sizable HR dataset from the website Kaggle [1], which has 2,000,000 records with 12 different characteristics, this study examines employee data. Two main business questions are addressed by the analytical pipeline: (1) what factors have the greatest impact on operational employee status; and (2) does salary serve as a dependable and structured metric across employee status. This study identifies systemic trends in the distribution of salary and performance evaluation through the use of classification models, together with data visualization techniques to forecast employee attrition [6] and data mining to predict performance [2].

II. RELATED WORK

The focus of predictive modeling in employee management has definitely moved towards machine learning frameworks. Comparative studies of the machine learning models have shown that the models with tree architecture, for example, Random Forest algorithms, outperform the traditional models in the case of the issues about an employee's status [17], [22]. Additionally, AI has created new avenues for strategic workforce planning in the HR management process [23]. Studies on sustainable HR shows that psychological well-being is one of the foundations in shaping the performance of an organization [7]. More recent challenges include work-life balance and burnout amongst the newer generation of workers further emphasize the need of achieving job satisfaction [3].

There are a number of recent studies on workplace transition during the pandemic that are showing a widening assessment gap between on-site and remote working. Hybrid working from home has proven to have a positive effect on retention without negatively impacting objective performance [4]. Pay is still a subject of debate and recent research into psychology has studied whether compensation is truly a motivator for well-being and performance or if there is a need to meet additional psychological needs [8]. To speed up these investigations, datasets for predictive modelling of attrition have been made available to the research community [18, 25]. This research is an extension of the known methodologies with a predictive, exploratory pipeline for assessing structural alignment of a large enterprise system.

III. DATASET AND DATA PREPARATION

A. DATASET SPECIFICATIONS

We worked with a rather big dataset called "HR Data MNC" [1], which contains entries for 2,000,000 employees spread across 12 columns. Numerical data (such as Salary_INR, Experience_Years, and Performance_Rating), category information (such as Department, Job_Title, Location, Status, and Work_Mode), a date field (Hire_Date), and a few identifier columns are all captured by these columns. A couple things jump out when you look at the stats up front. With an average of around 896,888 INR and a standard deviation of about 402,610 INR, salaries were all

over the place, indicating a wide range. The distribution of years of experience is quite uniform, ranging from 0 to 15. A well balanced distribution across the workforce is suggested by performance evaluations, which are based on a scale of 1 to 5 and do not significantly skew toward either extreme.

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The raw data had to be processed before any analysis could be started. There are 5 steps to get it into shape.

1. Three columns were dropped right away: Unnamed: 0, Employee_ID, and Full_Name. None of them carry any analytical value, so keeping them around would just add noise.
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5. Next, we included two interaction variables (Experience $\times$ Performance and Tenure $\times$ Performance), a consistent encoding for the country/location field, and Tenure_Years derived from Hire_Date to give the model additional signal.

IV. EXPLORATORY DATA ANALYSIS

A. UNIVARIATE DISTRIBUTIONS

The IT department comes out on top with close to 500,000 employees, ahead of Sales and Operations. More interesting is the spread across the Status categories. Active employees make up roughly 1.2 million records, Resigned employees account for around 400,000, and the Retired and Terminated groups are considerably smaller. That kind of imbalance is worth flagging early since it could affect classification performance down the line. Work mode, by contrast, sits at about 58% on-site and 42% remote, which is a much more even split.

B. BIVARIATE RELATIONSHIPS

Several things in the bivariate analysis didn't quite add up. Performance ratings looked nearly identical across all departments, which is unusual and suggests the variable doesn't carry much real variation. A box plot of Salary_INR against Performance_Rating backed this up: medians and

ranges overlapped heavily across all rating groups, so higher-rated employees weren't really earning more. Checking employee status against performance ratings showed the same thing, with Active, Resigned, Retired, and Terminated categories all turning up similar proportions at every rating level.

V. METHODOLOGY

A. MULTI-CLASS STATUS CLASSIFICATION

The models used to predict the status of the employee (Active, Resigned, Retired, or Terminated) were Random Forest Classifier and Decision Tree, which were trained using an 80/20 split. To avoid the model being influenced by any one variable, categorical variables were encoded using LabelEncoder and numeric variables were scaled using StandardScaler, which were applied prior to the model being fed into.

VI. RESULTS AND DISCUSSION

Two classification models were used for predicting the employee status as mentioned above, both trained on the same data set with the same preprocessing steps: Random Forest Classifier and Decision Tree model.

A. RESULTS FROM RANDOMFORESTCLASSIFIER

The overall predictive accuracy obtained with the Random Forest Classifier was 54%. Currently the predictive capacity of the model is restricted due to the high class imbalance [16]. The model performed well for most of the 'Active' class (F1 score of 0.70) but poorly for the remaining classes, with the 'Resigned' class having an F1 score of 0.20 and the 'Retired' and 'Terminated' classes being 0.05. The confusion matrix also shows that the classes that are regarded as the minority classes are misclassified with the majority class being 'Active'. The feature importance analysis shows that 'Salary_INR' is the most important feature contributing 95.8% and 'Experience_Years' contributing 2.4%, and 'Job_Title' and 'Performance_Rating' contributing 0.7% each.

B. RESULTS FROM DECISION TREE MODEL

The overall accuracy was slightly better with the Decision Tree Classifier (69%), but this is only said at the overall level and is misleading at the class level. As displayed in the classification report, the model for the class "Active" achieved a precision of 0.70, a recall of 0.97 and an F1 score of 0.81, which means that the model predicts "Active" in almost all of the cases. The other two classes, however, were much harder to classify, with "Resigned" getting an F1-score of only 0.02 and "Retired" and "Terminated" both 0.01, indicating poor overall performance. Additionally, the confusion matrix confirms that the majority of all true labels for each class, regardless of the class, were misclassified as "Resigned" or as various other classes that were not the true class, with the

diagonal (correct prediction) being surprisingly low for all classes other than "Active." This apparent increase in accuracy is further confirmed by the macro average F1-score of 0.22, which is not a true measure of discriminative learning but rather due to class imbalance.

The feature importance of the Decision Tree showed a more scattered and yet imbalanced distribution. Feature f5 accounted for 60.8% of importance, followed by f3 at 13.2%, f1 at 9.1%, f2 at 7.6%, f0 at 5.2%, and f4 at 4.1%. The dominance of a single feature is similar to the Random Forest and indicates that one variable (probably Salary_INR) continues to split the data in an inexplicable way.

VII. KEY FINDINGS AND INSIGHTS

A. SEVERE CLASS IMBALANCE

Since 70% of the data originates from 'Active' employees, both classification models predicted this label. The F1 scores obtained from Random Forest were 0.70, 0.20, 0.05, and 0.05 for 'Active', 'Resigned', 'Retired', and 'Terminated', respectively. Decision Tree produced a higher F1-Score for 'Active' with near zero for all minority classes, thereby revealing more bias towards the majority class.

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The most important feature is 'Salary_INR' with an importance of 95.8% in the Random Forest model and 61.0% in the Decision Tree model. As the same consistency holds across both models, it is clear that all other features indicate noise, but not actual predictors of employee status.

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The model heavily misclassifies 'Resigned', 'Retired', and 'Terminated' employees as 'Active', confirming the bias toward the majority class rather than genuine prediction.

VIII. RECOMMENDATIONS

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The 57% gap points to something an internal compensation audit could help explain. Wage entropy will be eliminated and labor cost will be optimized by establishing strict, transparent wage bands that are directly related to Experience_Years and functional departments [8], [24].

B. ANALYTICAL FEATURE ENRICHMENT

The HR data architecture has to include behavioral columns, such as manager effectiveness ratings, internal promotion interval, and sentiment scores from localized pulse surveys, in order to increase turnover classification accuracy over 54% [3],[7],[15].

IX. CONCLUSION

In this project, a large-scale enterprise HR dataset with two million records was subjected to an advanced data analytics and machine learning pipeline. By applying the tree-based classification, both with Random Forest and Decision Tree models, the misalignments in the structure are revealed. Due to its massive entropy, compensation significantly biases both classification models even though it was shown to be highly decentralized, with its variation irrespective of performance or experience. Random Forest outperformed Decision Tree with a higher accuracy for all employee status classes, whereas Decision Tree had a higher overall accuracy of 69%. Future models need to include features that better represent behaviors and class-balancing techniques as a means of creating a complete corporate attrition engine, even though the current models provide operational value in the identification of steady and active employees immediately [15], [16], [19].

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Abstract—This paper applies the study of data analytics to human resource management using a dataset representing employee records from a multinational corporation. This study investigates business questions such as factors influencing employee operational status and ultimately predicting employee attrition. A total of three machine learning types were used: multiple regression model (Random Forest, XGBoost, LightGBM, and Stacking) [21], Random Forest Classifier for employee status prediction [17], [22], and K-Means Clustering for unsupervised segmentation. The regression models reached a consistent R^2 score of 0.43. The Random Forest Classifier showed an accuracy of 54% with an F1-score of 0.70, driven by Salary INR with a feature importance at 95.8%. K-Means Clustering with $k = 2$ showed a split in the workforce based on work mode. These findings should provide insights to raise employee attrition within the company.

Keywords—HR Analytics, Employee Attrition, Random Forest Classification, K-Means Clustering, Feature Importance, Work Mode, Predictive Modeling.

I. INTRODUCTION

Human Resource (HR) analytics is a critical discipline in modern organizations. Transforming what was once a mundane operation into a strategic and data-driven decision engine [10], [11]. Large multinational corporations frequently face financial and operational loss due to employee turnovers due to expenses relating to recruitment and loss of organizational momentum [14]. Traditional HR practices rely on a more reactive framework, such as analyzing exit interviews. On the other hand, predictive data analytics allows organizations to identify turnover risks and evaluate the distribution of compensation, which should impact well for the organizational performance [9], [15]. However, the integration of HR analytics must also be able to navigate ethical challenges regarding employee data governance [5].

This study explores employee data using a large scale HR dataset sourced from the website Kaggle [1], containing 2,000,000 records with 12 distinct attributes. The analytical pipeline focuses on answering three primary business questions: (1) what factors most significantly influence operational employee status, (2) does salary function as a reliable and structured metric across employee status, and (3) how work modes influence performance evaluation and retention. By using regression, classification, and clustering models, helped with data visualization techniques to predict employee attrition [6] and data mining to predict performance [2], this paper finds systemic patterns in the distribution of compensation and performance evaluation.

II. RELATED WORK

Predictive modeling in employee management has very much shifted toward machine learning frameworks. Studies on comparative analysis of machine learning models confirm that tree-based architectures, such as Random Forest algorithms, have better performance over traditional regression models when handling employee status problems [17], [22]. Moreover, the integration of artificial intelligence into the HR management process has opened new roads for strategic workforce planning [23]. Studies on sustainable HR shows that psychological well-being is one of the foundations in shaping the performance of an organization [7]. More recent challenges include work-life balance and burnout amongst the newer generation of workers further emphasize the need of achieving job satisfaction [3].

To achieve this, unsupervised learning techniques like K-Means clustering are utilized to establish workforce segmentation, including cluster-based approaches for predicting turnover [20]. Recent studies about post-pandemic workplace transitions show an increasing deviation in how remote and on-site performance is evaluated. Researchers

have proven that hybrid working from home improves employee retention without damaging objective performance [4]. The debate regarding pay also remains as recent psychological studies explore whether compensation genuinely acts as a motivational mechanism for well-being and performance, or if its impact is dependent on other psychological needs as well [8]. Datasets for predictive attrition modelling have been made available to the research community to accelerate these investigations [18], [25]. This study builds on top of these established methodologies by applying a predictive and exploratory pipeline to assess structural alignment within a massive corporate dataset.

III. DATASET AND DATA PREPARATION

A. DATASET SPECIFICATIONS

For this project, we used the "HR Data MNC" dataset [1], which contains 2,000,000 employee records across 12 columns, covering numeric values such as Salary_INR, Experience_Years, and Performance_Rating, along with categorical values such as Department, Job_Title, Location, Status, Work_Mode, and Hire_Date, as well as some identifier columns.

Initial statistics reveal that salaries vary quite widely, with an average of around 896,888 INR and a standard deviation of roughly 402,610 INR. Experience years spread fairly evenly from 0 to 15, and performance ratings run on a 1–5 scale with no strong skew in either direction.

B. DATA PREPROCESSING PIPELINE

Before running any analysis, we cleaned and prepared the data through five main steps:

1. Dropping identifier columns. We removed 'Unnamed: 0', 'Employee_ID', and 'Full_Name', since they carry no analytical values..
2. Parsing the 'Hire_Date' column came in as a string, so we converted it to datetime64.
3. Checking data quality. We ran checks for both missing values and duplicates, fortunately we found out that neither was an issue. All 12 columns came back clean with zero nulls, and there were no duplicate rows.
4. Handling outliers. We applied the IQR method on the numeric columns, removing any values that fell outside the $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$ range.
5. Feature Engineering. To give the regression model more to work with, we created a few new features: 'Tenure_Years' (calculated from 'Hire_Date'), two interaction terms (Experience x Performance) and

(Tenure x Performance), and also encoded country/location as a standardized feature.

IV. EXPLORATORY DATA ANALYSIS

A. UNIVARIATE DISTRIBUTIONS

Organizational analysis shows that the IT Department is the largest department with around ~500,000 employees, followed by Sales and Operations. The target label, 'Status', shows a big imbalance where 'Active' contributes to around ~1,200,000 of the workforce, followed by 'Resigned' at around ~400,000 of the workforce, with 'Retired' and 'Terminated' occupying the other small percentages. Work modes show a somewhat stable split, with On-site accounting for around 58% and Remote accounting for 42% of the workforce.

B. BIVARIATE RELATIONSHIPS

The bivariate graph shows several inconsistencies. Performance ratings seem to be uniformly identical across all departments. A box plot of Salary_INR against Performance_Rating also shows an overlapping median and range, indicating that higher-rated employees do not seem to systematically receive superior compensation. Furthermore, tracking employee status across performance showed that all ratings are distributed proportionally.

V. METHODOLOGY

A. MULTI-MODEL REGRESSION FOR COMPENSATION ANALYTICS

To look at whether compensation is distributed fairly across employees, we ran four regression models: Random Forest Regressor, XGBoost, LightGBM, and a Stacked Ensemble [21]. Rather than relying solely on the original features, we expanded the feature set through engineering by adding Tenure, Location, and an interaction term between experience and performance (Exp_x_Perf) to give the models more context. Model performance was then compared using three metrics: R^2 , Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).

B. MULTI-CLASS STATUS CLASSIFICATION

For predicting employee status (Active, Resigned, Retired, or Terminated), we trained a Random Forest Classifier using an 80/20 train-test split [17], [22]. Before training, categorical columns were encoded with LabelEncoder, while numerical features were scaled using StandardScaler to prevent any single variable from dominating the model.

C. UNSUPERVISED WORKFORCE SEGMENTATION

To identify natural groupings within the workforce, we applied K-Means Clustering on four features: Experience_Years, Salary_INR, Performance_Rating, and

Work_Mode [20]. The optimal number of clusters was determined by plotting inertia curves with the Elbow Method and checking Silhouette Scores across k values from 2 to 7. Once the clusters were defined, we used Principal Component Analysis (PCA) to compress the multi-dimensional profiles down to two dimensions for easier visualization and interpretation.

VI. RESULTS AND DISCUSSION

A. REGRESSION METRICS

The Model evaluation revealed that regression architectures performed with remarkable uniformity across the board. In fact, that R^2 score achieved 0.43 this indicates that 57% of variance of 'Salary_INR' cannot be explained by the features such as experience, department, performance, and tenure. This statistical ceiling closely aligns with existing literature[8]. Which suggests that base pay mechanisms are often decentralized rather than solely dictated by meritocratic variables. The data generation shows highly randomized salary assignment or depends heavily on unmeasured variables, such as individual negotiations.

Table I: Regression Model Performance

Model	R^2 Score	MAE (INR)	RMSE (INR)
Random Forest	0.43	197,711	237,616
XGBoost	0.43	197,689	237,606
LightGBM	0.43	197,681	237,592
Stacking Ensemble	0.43	197,675	237,556

B. CLASSIFICATION PERFORMANCE AND FEATURE IMPORTANCE

The Random Forest Classifier attained an overall predictive accuracy of 54%. Nevertheless the model utility was heavily localized due to the majority class[16]. The model demonstrated a strong performance for the majority class with F1-score of 0.70, but dropped to 0.20 for 'Resigned' class and 0.05 for both 'Retired' and 'Terminated'. Confusion Matrix also reveals a significant misclassification of minority classes.

Furthermore, feature importance highlights that 'Salary_INR' has a big impact with 95.8%. On the other hand, 'Experience_years' contributed 2.4%, and both 'Job title' and 'Performance_Rating' contributed 0.7%. This indicates that salary_inr is a factor differentiating high-entropy rows,

rather than reflecting a valid causal pattern between compensation and firm attrition dynamics.[15][16]

C. WORKFORCE CLUSTERING PROFILES

As a direct result of the Elbow inertia curve and the maximum global silhouette score (0.275), the unsupervised clustering process succeeded at $K=2$. Interestingly, this clean delineation proved to be driven entirely by workplace mode.

Table II: K-Means Cluster Profiles ($k = 2$)

Cluster Label	Remote %	Mean Experience	Mean Salary (INR)	Mean Performance
Cluster 0	100%	5.0 years	829,126	3.0
Cluster 1	0% (On-site)	5.0 years	829,164	3.0

By achieving an absolute separation obtained by the K-means algorithm confirms that work modality operates as an independent structural axis. Specifically the numerical factors (experience_years, income, and quantitative performance_rating) are identical between two clusters. Furthermore, this statistic supports the findings of Bloom et al.[4], which assert that remote workmode does not in inherently harm performance. After that the automatic labeling of cluster 0 as 'low performance' and cluster 1 as 'high performance' as a systematic bias, rather than an objective deficiency in productivity among remote staff.

VII. KEY FINDINGS AND INSIGHTS

A. PARADOX OF SALARY DOMINANCE

Salary_INR is very independent in regression modeling ($R^2 = 0.43$), despite giving 95.8% of the status classification framework. This demonstrates that salary is not structurally aligned with tenure or performance and is distributed with a large degree of volatility [8],[24].

B. MODAL BIFURCATION AND EVALUATION BIAS

The company dataset separates remote status, as demonstrated by the unsupervised segregation of 100% of remote employees into a distinct cluster. Any administrative categorization that labels remote employees as bad performers given a similar quantitative performance average ($\mu = 3.0$) indicates systemic presenteeism bias [4],[6].

C. ATTRITION DATA DEFICIT

The lack of ability to forecast active turnover (Resigned $F1 = 0.20$) shows that structural characteristics are not enough on their own. Behavioral telemetry, including role satisfaction score, engagement indices, and management sentiment, is necessary for attrition prediction [3],[7],[19].

D. ATTRITION DATA DEFICIT

The approach may be used right away to map out the basic employees segment with minimal immediate attrition risk, as demonstrated by the strong identification of steady personnel (Active $F1 = 0.70$) [9].

VIII. RECOMMENDATIONS

A. INSTITUTIONAL COMPENSATION STANDARDIZATION

To address the 57% wage disparity that cannot be explained, the company should conduct an internal compensation audit. Wage entropy will be eliminated and labor cost will be optimized by establishing strict, transparent wage bands that are directly related to Experience_Years and functional departments [8], [24].

B. MODERNIZATION OF REMOTE PERFORMANCE METRICS

To remove any presenteeism biases, management has to re-engineer remote work performance frameworks. Remote clusters will be equally aligned with key company performance objectives by substituting objective, output based KPIs for physical presence tracking [4].

C. ANALYTICAL FEATURE ENRICHMENT

The HR data architecture has to include behavioral columns, such as manager effectiveness ratings, internal promotion interval, and sentiment scores from localized pulse surveys, in order to increase turnover classification accuracy over 54% [3],[7],[15].

D. PRECISION RETENTION DEPLOYMENT

The K-Means structure should be used by the HR to protect valuable assets. In order to reduce replacement costs, Cluster 1 on-site employees who maintain optimal objective performance should be given priority for accelerated career advancement and specialized rewards [20],[23].

IX. CONCLUSION

In this project, a large-scale enterprise HR dataset with two million records was subjected to an advanced data analytics and machine learning pipeline. Several structural misalignments within the organization have been revealed by unsupervised clustering, tree-based classification, and multi-model regression. Due to its massive entropy, compensation significantly biases classification models even though it was shown to be highly decentralized, with 57% of salary variation irrespective of performance or experience

[21]. Additionally, a complete operational divide between remote and on-site employees was shown by K-Means clustering, indicating a structural vulnerability to proximity bias in performance management [4], [20]. In order to build a really comprehensive corporate attrition engine, future iterations must prioritize behavioral feature enrichment and class-balancing strategies, even if the existing model offers immediate operational value in identifying steady, active personnel ($F1 = 0.70$) [15], [16], [19].

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